

TRACK 2: DEVELOPER AGENTS

Test Case Generator Bot

Aryan Saxena

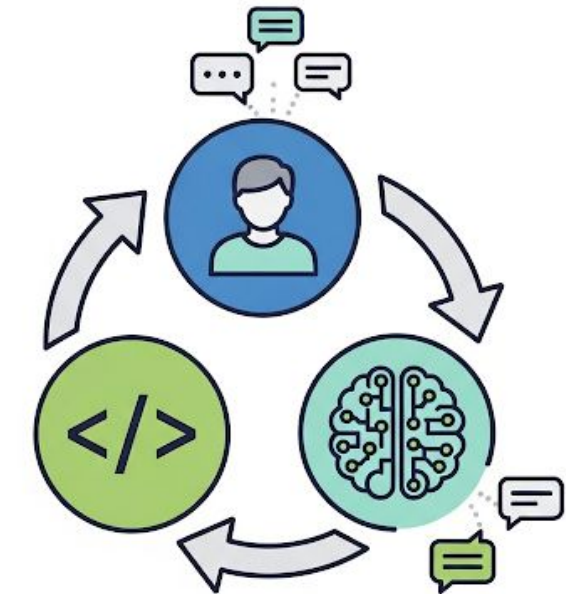
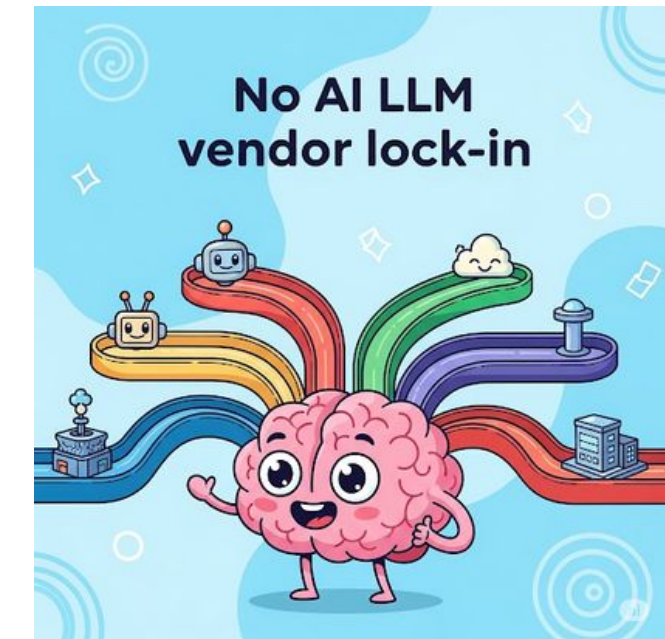
PROBLEM STATEMENT

- Modern software projects require robust, automated testing to ensure code quality and reliability.
- However, writing comprehensive test cases is time consuming and often neglected, especially for legacy code or rapid prototyping.
- Many developers lack the expertise or time to create effective tests for every function, leading to bugs, regressions, and reduced confidence in code changes.



SOLUTION

- **Voylla** a Python package and CLI tool for automated test generation.
- Supports Python, Java, and JavaScript (**more languages can be added easily**)
- Leverages advanced AI models (OpenAI, Anthropic) to generate:
 - Unit, integration, and edge test cases
 - Coverage insights and detection of edge conditions/performance risks
- Interprets code flow and logic for smarter, more relevant test generation.
- Interactive refinement: users can review and improve generated tests conversationally.
- True Python package (Published !): install via pip, use your own API keys (**zero cost privacy friendly no vendor lock in**)



Interactive refinement

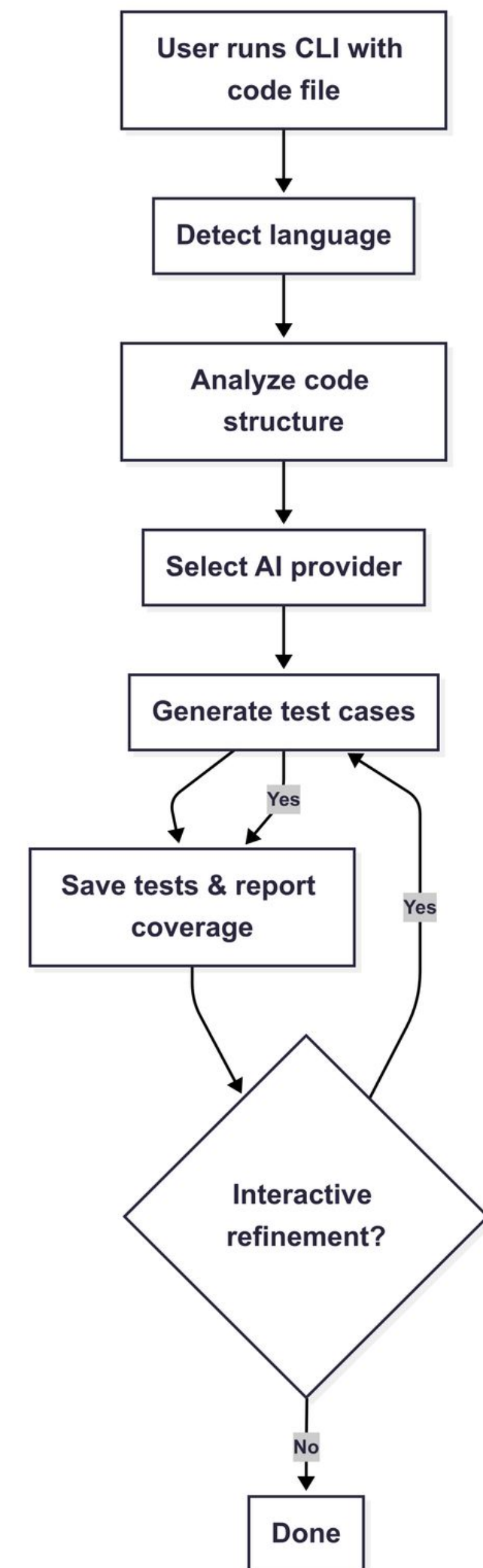


True Python Package
(Published!)

[!\[\]\(003082e50e3009141f59bd5df831749f_img.jpg\) ***Published Link!***](#)

APPROACH

- **User Input:** The user submits a code file via the CLI
- **Language Detection:** Voylla auto detects the language if not specified.
- **Code Analysis:** The analyzer parses the code, extracting functions, classes, and control flow to understand logic and edge cases.
- **AI Provider Selection:** The configuration manager selects the best available AI provider
- **Test Generation:** The generator uses the AI provider to create comprehensive test cases, including unit, integration, and edge tests, and highlights performance risks.
- **Output & Coverage:** Generated tests are saved to the specified directory, and a coverage report is provided.
- **Interactive Refinement:** Users can interactively review, refine, and regenerate test cases in a conversational loop.



TECH STACKS AND FRAMEWORKS USED

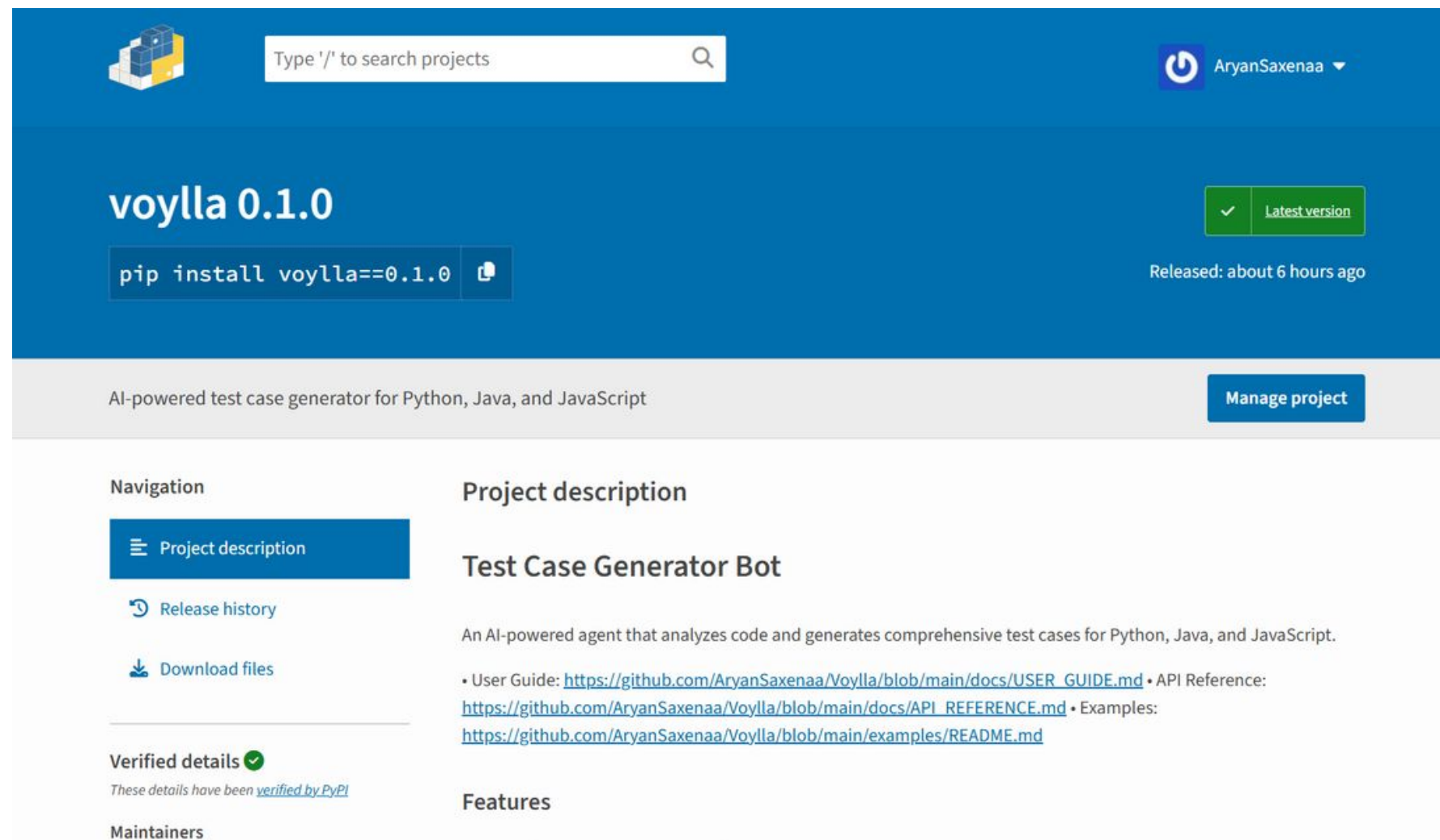
- **Python 3.10+:** The essential programming language for the tool.
- **Click:** Utilized for creating a user-friendly command-line interface.
- **Rich:** Employed for delivering colored, formatted terminal output and panels.
- **PyYAML:** Used for reading configuration files in YAML format.
- **python-dotenv:** Facilitates loading environment variables from .env files (such as API keys and configurations).
- **OpenAI & Anthropic SDKs:** Integrate with AI models to generate test cases.
- **tree-sitter & language bindings:** Allow for parsing and analyzing code in Python, Java, and JavaScript.
- **pytest:** For writing and running automated tests on the tool itself.
- **setuptools, wheel:** Essential for packaging and distributing the project.

Each component has been selected for its reliability, strong community support, and suitability for both educational and production purposes. The modular architecture supports

WHY YOU CHOSE THIS PROBLEM STATEMENT?

- I chose this problem statement because of my prior experience as a Fellow at Keploy, an open source testing tool focused on automating test case generation and improving developer productivity.
- During that time, I gained a strong understanding of how automated test generation works, its challenges, and its potential to save significant development time.
- This background gave me both the technical insight and the motivation to explore building a test generator bot.
- I see this as an opportunity to apply my past learnings, experiment with AI-assisted approaches, and create a solution that can help developers write better tests more efficiently.

SCREENSHOTS



Install Voylla Via Simple Pip command and Use it via interactive CLI

```
91730@Aurum MINGW64 ~/Documents/Hackathons/Agentforce video/Billing_system
$ source "/C/Users/91730/Documents/Hackathons/Agentforce video/Billing_system/.venv/Scripts/activate"
(.venv)
91730@Aurum MINGW64 ~/Documents/Hackathons/Agentforce video/Billing_system
$ pip install voylla
Requirement already satisfied: voylla in c:\users\91730\documents\hackathons\agentforce video\billing_
Requirement already satisfied: click>=8.0.0 in c:\users\91730\documents\hackathons\agentforce video\bi
```


91730@Aurum MINGW64 ~/Documents/Hackathons/Agentforce video/Billing_system
\$ voylla --file biling_system.py --output test_output

Test Case Generator Bot

Analyzing code and generating intelligent test cases...

AI Enhancement: Disabled (set OPENAI_API_KEY or ANTHROPIC_API_KEY, or create a .env)
Analyzing biling_system.py...
Analyzing code structure and flow...
Code Analysis Summary

Metric	Value
Language	python
Functions Found	8
Classes Found	1
Complexity Score	5
Edge Cases Detected	5
Performance Risks	0

⚠ Detected Edge Cases:

- Division by zero
- Index out of bounds
- Invalid type conversion
- File not found
- Empty string splitting

✍ Generating test cases...
🤖 Enhancing tests with AI insights...
✓ Generated test_output\test___init__.py
✓ Generated test_output\test_total.py
✓ Generated test_output\test_welcome_bill.py
✓ Generated test_output\test_bill_area.py
✓ Generated test_output\test_save_bill.py
✓ Generated test_output\test_find_bill.py
✓ Generated test_output\test_clear_data.py
✓ Generated test_output\test_exit_app.py
✓ Generated 34 test cases
✓ Coverage: 100.0%
(.venv)

Metric	Value
Language	python
Functions Found	8
Classes Found	1
Complexity Score	5
Edge Cases Detected	5
Performance Risks	0

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- Division by zero
- Index out of bounds
- Invalid type conversion
- File not found
- Empty string splitting

✍ Generating test cases...
🤖 Enhancing tests with AI insights...
✓ Generated test_output\test___init__.py
✓ Generated test_output\test_total.py

USE CLI COMMANDS TO USE VOYLLA

USE INTERACT COMMAND TO USE MULTIPLE OPTIONS TO ENHANCE THE TEST CASE AS PER YOU WISH !

(.venv)
91730@Aurum MINGW64 ~/Documents/Hackathons/Agentforce
\$ voylla --file biling_system.py --interactive

Test Case Generator Bot

Analyzing code and generating intelligent test cases...

25	test_find_bill_edge_index_error	edge	find_bill
26	test_find_bill_edge_file_not_found	edge	find_bill
27	test_clear_data_basic	unit	clear_data
28	test_clear_data_edge_division_by_zero	edge	clear_data
29	test_clear_data_edge_index_error	edge	clear_data
30	test_clear_data_edge_file_not_found	edge	clear_data
31	test_exit_app_basic	unit	exit_app
32	test_exit_app_edge_division_by_zero	edge	exit_app
33	test_exit_app_edge_index_error	edge	exit_app
34	test_exit_app_edge_file_not_found	edge	exit_app

What would you like to do? [view/edit/add/remove/enhance/save/quit] (quit): rem
Enter test ID to remove: 33
✓ Removed test: test_exit_app_edge_index_error
Generated Test Cases